

CHE 230: Molecular Systems in Chemical Engineering Homework 1

Due: 02/11/25

1. Graphite and Diamond are chemically identical but different. Explain why and give 2 other examples of materials that exhibit a similar phenomenon. (4 points)

Carbon is an example of allotropy, two materials with the same chemical formula, but different structures and different properties. Diamond and graphite are both crystal carbon structures. Diamond is a crystal in which each carbon atom is bonded covalently to four other carbons through sp^3 hybridized orbitals. This makes the structure very strong. Graphite is formed by layered sheets of carbon, each atom bonded to three other atoms with sp^2 . Because each sheet is attracted to the others only by weaker secondary bonding, the structure is more fragile and layers can be rubbed off as in a pencil.

Other examples of allotropy are tin and iron. Below 13.2 C, white tin turns into gray tin, which has a different crystal structure and is more brittle. Iron exhibits a similar phenomenon at high temperature. It changes from a BCC crystal structure to FCC at temperatures above 912 C.

2. What is the main difference between ionic, covalent, and metallic bonds? Discuss and draw a diagram to explain. (6 points)

In ionic bonds, electrons are transferred from one atom to another, and the resulting positive and negative ions are attracted to each other by colombic forces. This occurs between metallic and non-metallic, where the difference in electronegativity between both atoms is large. Ionic bonds are non directional. Each positive ion will be bonded to several negative ions and vice versa. Bonding energy is large for ionic compounds.

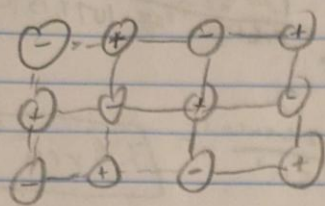
In covalent bonds, electrons are shared between atoms. Each atom in a covalent bond contributes at least one electron to the bond. The sharing of electrons allows atoms to fill their valence shell and become more stable. Covalent bonding occurs between atoms that do not have a great difference in electronegativity. Covalent bonding is directional, the atoms must be bonded in a specific arrangement.

In metallic bonding, atoms are arranged in a crystal structure and electrons are able to pass freely between them. This "sea" of electrons makes metals good conductors of heat and electricity.

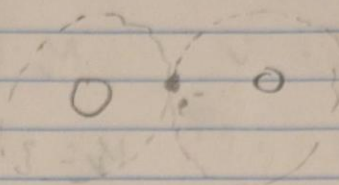
See next page for diagrams

2.

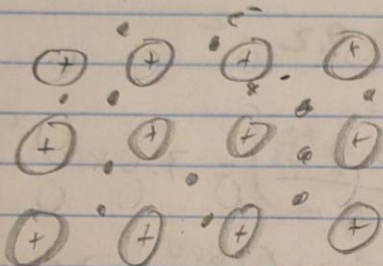
Ionic



Covalent

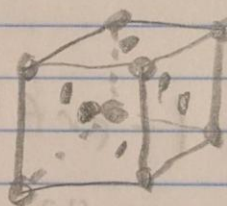


Metallic



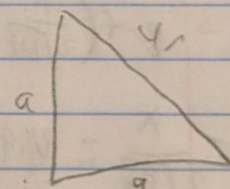
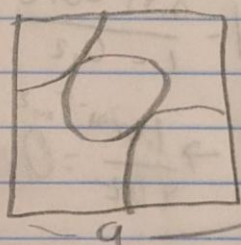
$$3a. APF = \frac{V_{atoms}}{V_{cell}}$$

$$FCC = \frac{1}{8} \cdot 8 + \frac{1}{2} \cdot 6 = 4 \text{ atoms/cell}$$



$$2a^2 = 16r^2 \quad r = 0.13 \text{ nm}$$

$$a = \sqrt{8}r \quad a = 0.404 \text{ nm}$$



$$V_{cell} = a^3 = 0.0662 \text{ nm}^3$$

$$V_{atoms} = 4 \cdot \frac{4}{3} \pi r^3 = 0.0122 \text{ nm}^3$$

$$APF = \frac{4 \cdot 0.0122 \text{ nm}^3}{0.0662 \text{ nm}^3} = \boxed{0.74}$$

$$b \text{ Al: } \frac{26.98 \text{ g}}{\text{mol}} \times \frac{1 \text{ mol}}{6.022 \times 10^{23} \text{ atoms}} \times \frac{4 \text{ atoms}}{0.0662 \text{ nm}^3} \times \left(\frac{10^7 \text{ nm}}{1 \text{ cm}} \right)^3 = \boxed{2.71 \frac{\text{g}}{\text{cm}^3}}$$

This is very close to actual Al density of 2.70 g/cm^3 . Real world density is affected by defects in the crystal system.

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4. You are a process engineering intern at a pharmaceutical firm developing a high-potency Active Pharmaceutical Ingredient (API) for pancreatic cancer treatment. During a routine characterization of the batch using X-ray diffraction (XRD) and microscopy, you identify significant **lattice defects** within the crystal samples.

- a. Should you be concerned about the presence of these defects regarding the final drug product? Justify your answer by discussing how defects can impact the **solubility** and **stability** of a pharmaceutical crystal. (4 points)

We should be concerned about the defects. Excessive defects in the pharmaceutical crystal could cause the solubility to become unpredictable. For example, if the defect is a vacancy, few bonds would need to be broken in order for the crystal to dissolve, and the compound would become more soluble. Stability is also affected, as defects can diffuse throughout the crystal, causing the ingredient to change its properties over time. Manufacturing consistency is critical, especially in the pharmaceutical industry.

- b. Identify **two** specific molecular or environmental factors during the crystallization process that could have led to these defects (4 points)

This could have been caused by improper control of temperature, for example the temperature could have been too high or the compound could have been cooled too quickly.

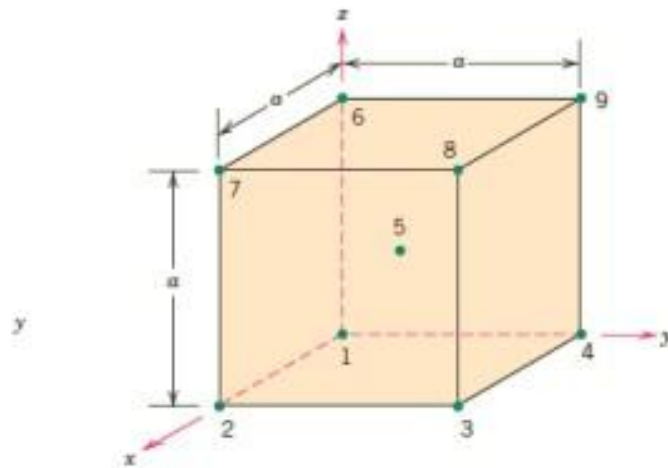
It could also have been caused by impurities that became included in the lattice.

- c. Identify **two industrial processes** in which crystal defects are critical to the product's function. (2 points)

Semiconductors use defects to modify and control electrical conductivity. Catalysts use defects to provide a surface for other reactions to occur.

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3. Write the coordinates for each of the points in the BCC (5 Points)



1. 000
2. 100
3. 110
4. 010
5. 0.5 0.5 0.5
6. 001
7. 101
8. 111
9. 011

✓ 9

$$800^{\circ}\text{C} = 1073.15 \text{ K}$$

$$6. \quad N_A = \frac{q_s}{\text{cm}^2} \times \frac{1 \text{ mol}}{107.9 \text{ g}} \times \frac{6.022 \times 10^{23} \text{ atoms}}{\text{mol}} = 5.7 \times 10^{22} \text{ atoms}$$

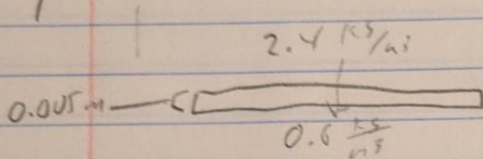
$$N_A + N_V = N_C$$

$$N_V = N_C - N_A$$

$$\frac{N_V}{N_A} = 1 - \frac{N_A}{N_C}$$

$$N_V = 3.6 \times 10^{17} \frac{\text{atoms}}{\text{cm}^3} = \boxed{3.6 \times 10^{17} \frac{\text{atoms}}{\text{cm}^3}}$$

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$$A = 0.2 \text{ m}^2$$

$$\tau = -D \frac{dC}{dx} = -10^{-8} \frac{\text{m}^2}{\text{s}} \left(\frac{0.6 - 2.4}{0.005 \text{ m}} \right) = 3.6 \times 10^{-1} \frac{\text{kg}}{\text{m}^2 \text{ s}}$$

$$\dot{m} = \tau A = (3.6 \times 10^{-1} \frac{\text{kg}}{\text{m}^2 \text{ s}}) (0.2 \text{ m}^2) \left(\frac{3600 \text{ s}}{\text{hr}} \right) = \boxed{2.592 \times 10^{-3} \frac{\text{kg}}{\text{hr}}}$$

$$8. a \quad \frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf} \left(\frac{x}{2\sqrt{Dt}} \right)$$

$$C_x = 0.35$$

$$C_0 = 0.2$$

$$C_s = 1$$

$$\text{erf} \left(\frac{x}{2\sqrt{Dt}} \right) = 1 - \frac{0.35 - 0.2}{1 - 0.2} = 0.8125$$

$$t = 49.54 = 178200 \text{ s}$$

$$x = 0.004 \text{ m}$$

$$\frac{x}{2\sqrt{Dt}} = 0.932 \rightarrow \frac{1.6 \times 10^{-5} \text{ m}^2}{4Dt} = 0.8686$$

$$\boxed{D = 2.6 \times 10^{-11} \frac{\text{m}^2}{\text{s}}}$$

$$b \quad D = D_0 e^{-\frac{Q_d}{RT}} \rightarrow \ln \frac{D}{D_0} = -\frac{Q_d}{RT} \rightarrow T = \frac{-Q_d}{R \ln \frac{D}{D_0}}$$

$$\boxed{T = 1300 \text{ K}}$$